Gym ERP Blueprint

BACKEND ENGINE & TECHNICAL SPECIFICATIONS: PHYSICAL ASSESSMENT & NUTRITION MODULE

This technical document provides a production-ready system specification for implementing a comprehensive **Physical Assessment and Nutrition Module** within a Gym ERP from scratch. It defines the structural dependencies, precise algorithmic logic using standard gym metrics (no laboratory tests required), database schema definitions, and visual dashboard layouts.

1. Architectural System Flow & Dependencies

To avoid mathematical reference errors, calculations must execute in a strict sequential pipeline. Several metrics depend entirely on the outputs calculated in preceding stages:

1. **Onboarding Inputs:** System collects basic demographic data (Age, Gender) and primary physiological baselines (Weight, Height, Resting Heart Rate).
2. **Base Metrics Matrix:** The system computes standalone benchmarks: Body Mass Index (BMI), Basal Metabolic Rate (BMR), and Ideal Body Weight (IBW).
3. **Circumference Processing:** Utilizing tape measurements, the engine computes the Waist-to-Hip Ratio (WHR) and derives the Body Fat Percentage (BF%) via non-invasive regression formulas.
4. **Body Composition Extraction:** Lean Body Mass (LBM) is calculated directly by combining the total Body Weight and the calculated Body Fat Percentage.
5. **Energy Expenditure Matrix:** Total Daily Energy Expenditure (TDEE) and target daily caloric boundaries are established based on the member's lifestyle multipliers and fitness targets.
6. **Nutritional Allocation & Target Cardiovascular Zones:** The macro split (Protein, Fats, Carbs in grams) is dynamically configured, and target heart rate brackets are derived.

2. Database Schema Specification

Your development team must establish a relational database table (e.g., `member_assessments`) capable of collecting and storing the following structured attributes for every progress log entry:

Field Name	Data Type	Validation / Constraints	UI / Physical Source Context
member_id	Foreign Key	Must resolve to primary keys of main member table.	Automated session state context.
assessment_date	Date	Default: Current Timestamp.	System generated entry record date.
gender	Enum	`['male', 'female']`	Retrieved from standard member metadata.
age	Integer	Minimum: 10 | Maximum: 100 (years)	Calculated dynamically or profile input.
weight_kg	Decimal (5,2)	Minimum: 30.00 | Maximum: 250.00	Standard medical weight scale reading.
height_cm	Decimal (5,2)	Minimum: 100.00 | Maximum: 250.00	Stadiometer or height-rod reading.
neck_cm	Decimal (4,1)	Minimum: 20.0 | Maximum: 60.0	Anthropometric tape line (just under larynx).
waist_cm	Decimal (4,1)	Minimum: 40.0 | Maximum: 180.0	Anthropometric tape line (horizontal at navel level).
hip_cm	Decimal (4,1)	Nullable for Men; Required for Women.	Anthropometric tape line (widest gluteal span).
resting_hr	Integer	Minimum: 40 | Maximum: 120 (BPM)	Manual pulse count or pulse oximeter device.
activity_level	Enum	`['sedentary', 'light', 'moderate', 'active']`	Dropdown option representing weekly baseline activity.
fitness_goal	Enum	`['fat_loss', 'maintenance', 'muscle_gain']`	Dropdown option dictating energy delta application.

3. Algorithmic Processing Engine Equations

The backend logic processing service must run the data attributes through the following deterministic pipelines sequentially, mapping internal variable names appropriately:

> **Metric 1: Body Mass Index (BMI)**
>
> $$BMI = weight\_kg / (height\_cm / 100)^2$$
>
> **System Risk Classification Status Flags:**
>
> - $BMI < 18.5$ → Flag Status: `"Underweight"` (Triggers safety warnings for hypertrophy priority)
> - $BMI \geq 18.5$ AND $BMI < 25.0$ → Flag Status: `"Normal"`
> - $BMI \geq 25.0$ AND $BMI < 30.0$ → Flag Status: `"Overweight"`
> - $BMI \geq 30.0$ → Flag Status: `"Obese"` (Triggers high joint-impact training alerts)

Metric 2: Basal Metabolic Rate (BMR) - Mifflin-St Jeor Engine

Evaluate conditionally based on `gender` attribute:

$$\text{If gender == 'male': BMR} = (10 \times \text{weight\_kg}) + (6.25 \times \text{height\_cm}) - (5 \times \text{age}) + 5$$

$$\text{If gender == 'female': BMR} = (10 \times \text{weight\_kg}) + (6.25 \times \text{height\_cm}) - (5 \times \text{age}) - 161$$

Significance: Establishes the metabolic floor. Absolute minimum caloric energy required to sustain life at total rest. Critical for system boundary controls to ensure active nutrition plans never fall below this value.

Metric 3: Total Daily Energy Expenditure (TDEE)

Map `activity_level` variable to hardcoded numeric scale multipliers:

```
'sedentary' → 1.200 | 'light' → 1.375 | 'moderate' → 1.550 | 'active' → 1.725
```

$$\text{TDEE} = \text{BMR} \times \text{Multiplier}$$

Significance: Establishes equilibrium energy balance (Maintenance calories). Eating exactly this number maintains current weight.

Metric 4: Target Calorie Budgeting

Evaluate variable `fitness_goal` against the calculated TDEE baseline to inject adjustments:

$$\text{If fitness\_goal == 'fat\_loss': Target Calories} = \text{TDEE} - 500$$

$$\text{If fitness\_goal == 'maintenance': Target Calories} = \text{TDEE}$$

$$\text{If fitness\_goal == 'muscle\_gain': Target Calories} = \text{TDEE} + 350$$

Metric 5: Body Fat Percentage (BF%) - US Navy Circumference Methodology

Compute using base-10 logarithmic scaling functions (`log10`) on tape parameters:

$$\text{If gender == 'male': BF\%} = 86.010 \times \log_{10}(\text{waist\_cm} - \text{neck\_cm}) - 70.041 \times \log_{10}(\text{height\_cm}) + 36.76$$

$$\text{If gender == 'female': BF\%} = 163.205 \times \log_{10}(\text{waist\_cm} + \text{hip\_cm} - \text{neck\_cm}) - 97.684 \times \log_{10}(\text{height\_cm}) - 78.387$$

Significance: Differentiates fat tissue from lean skeletal mass. Eliminates inaccuracies caused by standard weight metrics in highly muscular or detrained individuals.

Metric 6: Lean Body Mass (LBM)

$$\text{LBM} = \text{weight\_kg} \times (1 - (\text{BF\%} / 100))$$

Significance: Determines total non-adipose weight (muscles, skeletal system, organs). Acts as the foundational variable dictating systemic protein breakdown metrics.

Metric 7: Macronutrient Allocation Split Engine

Process sequentially based on total `Target Calories` and structural calculated `LBM` parameters:

1. **Protein Split:** Allocate 2.2 grams per kilogram of LBM.

   $$Protein\_Grams = LBM \times 2.2$$

   $$Protein\_Calories = Protein\_Grams \times 4$$

2. **Lipids/Fat Split:** Allocate exactly 25% of the total Target Calories daily allocation.

   $$Fat\_Calories = Target\ Calories \times 0.25$$

   $$Fat\_Grams = Fat\_Calories / 9$$

3. **Carbohydrates Split:** Allocate the remaining mathematical balances to energy inputs.

   $$Carb\_Calories = Target\ Calories - (Protein\_Calories + Fat\_Calories)$$

   $$Carb\_Grams = Carb\_Calories / 4$$

Metric 8: Ideal Body Weight (IBW) - Devine Clinical Standard

$$Inches\_Over\_60 = ((height\_cm / 2.54) - 60)$$

$$If\ gender == 'male':\ IBW = 50.0 + (2.3 \times Inches\_Over\_60)$$

$$If\ gender == 'female':\ IBW = 45.5 + (2.3 \times Inches\_Over\_60)$$

Significance: Establishes an objective healthy weight baseline normalized against structural skeletal scale lengths. Grounding point for expectations during goal setting.

Metric 9: Waist-to-Hip Ratio (WHR)

$$If\ gender == 'male'\ AND\ hip\_cm == Null:\ WHR = Null$$

$$Else:\ WHR = waist\_cm / hip\_cm$$

Significance: Direct diagnostic metric flag for visceral fat clustering. Values > 0.90 for men or > 0.85 for women indicate apple-shaped fat distribution and elevated cardiovascular health risks.

Metric 10: Target Heart Rate Training Zones (Karvonen Method)

$$Max\_HR = 220 - age$$

$$HR\_Reserve\ (HRR) = Max\_HR - resting\_hr$$

Target Aerobic Intensity Brackets:

- Fat Burn Bounds Low (60% exertion): `(HRR × 0.60) + resting_hr`
- Fat Burn Bounds High (70% exertion): `(HRR × 0.70) + resting_hr`
- Cardio Endurance Range (70%-80% exertion): `From Fat Burn High Target to ((HRR × 0.80) + resting_hr)`

Developer Technical Note: Ensure proper input parameter constraints exist within the codebase to prevent divisions by zero or negative logarithmic parameters. In instances where male data inputs exclude `hip_cm`, handle the `WHR` variable gracefully by outputting `Null` or `N/A` on the user interface screens rather than fracturing calculation execution blocks.

4. UI/UX Dashboard Layout & Wireframe Component Maps

Instruct your frontend engineers to divide the compiled data calculations across 4 clean visual interface cards on the member evaluation viewport:

COMPONENT A: VITAL STRUCTURAL COMPOSITION BLOCK

Provides an immediate overview of body structure indicators. Muscle-to-fat changes must take visual priority over pure weight changes.

- **Primary Large Numeric Displays:** Total Scale Weight (`weight_kg`), calculated Body Fat percentage (`BF%`), and total structural Lean Body Mass (`LBM`).
- **Context Label Indicators:** Display calculated `BMI` values accompanied by color-coded indicator badges representing risk levels (Green: Normal, Amber: Overweight, Red: Obese) alongside `WHR` results.

COMPONENT B: ENERGY THRESHOLD ANALYTICS ENGINE

Highlights systemic energy baselines, defining the member's daily metabolic boundary limits clearly.

- **Primary Base Value:** Display `BMR` as the absolute structural minimum daily caloric floor line.
- **Actionable Threshold Data:** Display `TDEE` explicitly labeled as *"Maintenance Calories (Equilibrium Balance)"*, and frame the calculated `Target Calories` as a primary metric highlighted as the *"Active Daily Energy Budget Allocation"*.

COMPONENT C: MACRONUTRIENT MACRO BLUEPRINT PANEL

Converts target energy allocations into real-world daily nutritional goals for the client.

- **Visual Component Display:** Use a segmented progress bar or structural chart representing total daily energy allocation percentages.
- **Data Point Fields:**
  - **Protein:** Grams calculated displayed clearly, captioned with: *"Target allocation for skeletal muscle structure preservation/growth."*
  - **Carbohydrates:** Grams calculated, captioned with: *"Daily baseline glycolytic performance energy."*
  - **Fats:** Grams calculated, captioned with: *"Essential lipid intake block supporting foundational hormonal health and joint longevity."*

COMPONENT D: PROGRAMMATIC CARDIOVASCULAR TARGET GUIDANCE CARD

Translates heart rate formulas into clear parameters that can be easily applied to standard gym equipment interfaces (treadmills, spin bikes, ellipticals).

- **Automated Advisory Text Field:** Generate structured guidance copy block layout matching: *"Maintain working heart rate activity outputs bounded between **[Fat Burn Low]** BPM and **[Fat Burn High]** BPM during continuous metabolic conditioning routines to maximize fat oxidation efficiently. Transition into the aerobic training bracket of **[Fat Burn High]** to **[Cardio High]** BPM to increase cardiovascular stamina and threshold performance parameters."*